dissolved into a swarming cloud of probabilities, which the equations still manage to describe.

Our universe could thus be the result of the collapse of a previous contracting universe passing across a quantum phase, where space and time are dissolved into probabilities.

The word ‘universe’ becomes ambiguous. If, by ‘universe’, we mean ‘all that there is’, then, by definition, there cannot be a second universe. But the word ‘universe’ has assumed another meaning in cosmology: it refers to the spacetime continuum that we see directly around us, filled with galaxies the geometry and history of which we observe. There is no reason to be certain that, in *this* sense, this universe is the only one in existence. We can reconstruct the past up to the time when, as in the image by John Wheeler, the spatiotemporal continuum breaks up like sea foam and fragments into a quantum cloud of probabilities, and there is no reason to discard the possibility that beyond this hot foam there could not be another spatiotemporal continuum, similar to the one which we perceive around us.

The probability for a universe to cross the phase of the Big Bounce, passing from contraction to expansion, can be computed using the techniques described in the preceding chapter: the spacetime boxes. Calculations are made using spinfoams that connect the contracting universe with the expanding one.

All of this is still at an exploratory stage, but what is remarkable in this story is that today we have equations with which to try to describe these events. We are beginning to cast the first few cautious glances, for the moment only theoretically, beyond the Big Bang.